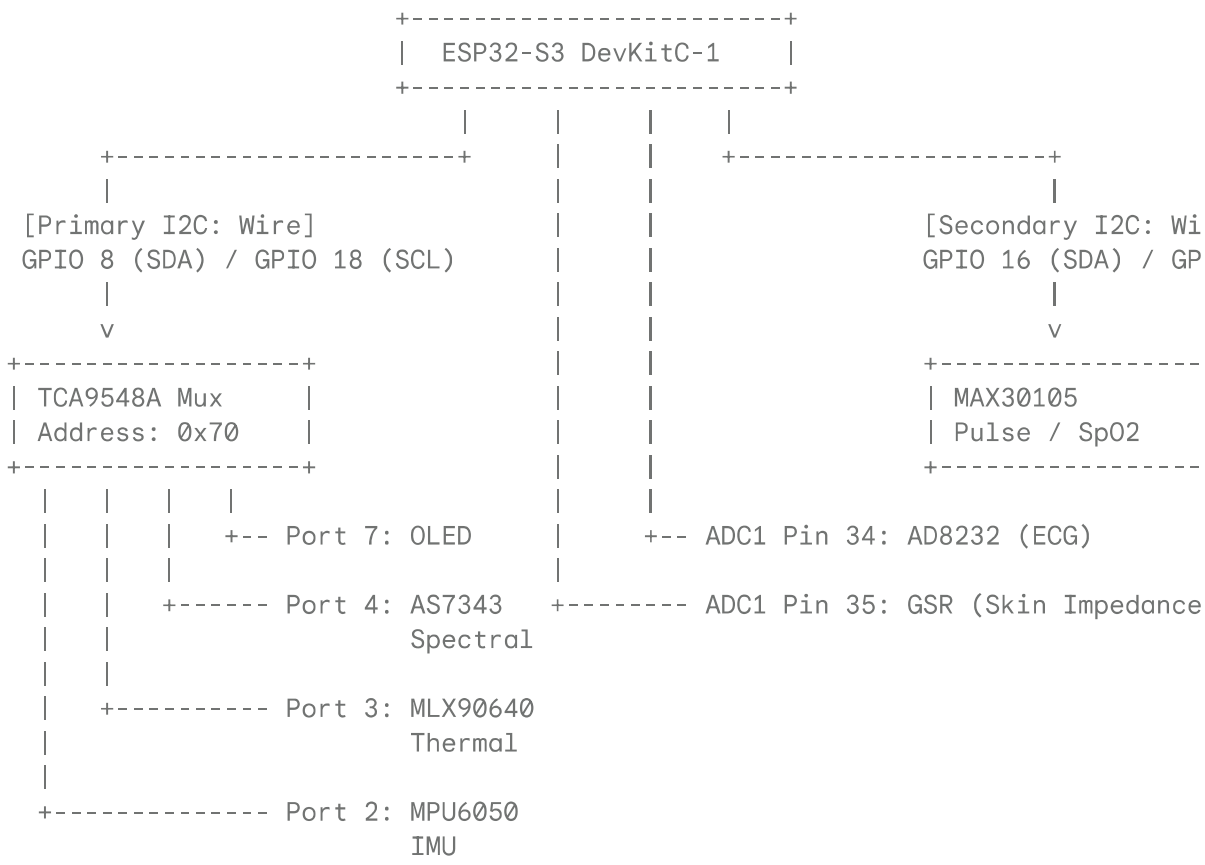
Smart-ASHA: Pinout, Wiring & Prototype Setup Guide

This document provides the complete hardware wiring scheme, multiplexer mapping, and electrical layout for the Smart-ASHA ESP32-S3 multi-modal diagnostic system.

1. System Architecture Diagram



2. Master Pin Connection Matrix

Subsystem / Sensor	Sensor Pin	ESP32-S3 Pin	Bus / Interface	Notes
TCA9548A Multiplexer	VCC	3.3V	Power Rail	Main bus multiplexer (Address: `0x70`)
	GND	GND	Common Ground	Common system ground
	SDA	GPIO 8	`Wire` (Primary I2C)	Clock speed set to $50\ \text{kHz}$

	SCL	GPIO 18	`Wire` (Primary I2C)	Clock speed set to $50\ \mathrm{kHz}$
	RST	3.3V	Logic High	Pulled high to prevent reset
MAX30105 Pulse/SpO2	VCC	3.3V	Power Rail	Completely isolated from TCA9548A
	GND	GND	Common Ground	
	SDA	GPIO 16	`Wire1` (Secondary)	Independent bus to prevent contention
	SCL	GPIO 17	`Wire1` (Secondary)	Independent bus to prevent contention
AD8232 ECG Sensor	3.3V	3.3V	Clean Power Rail	Keep away from switching noise
	GND	GND	Common Ground	
	OUTPUT	GPIO 34	Analog (ADC1_CH6)	Safe ADC1 channel (no Wi-Fi conflict)
	LO+	*Unused / Opt*	Digital In	Lead-off detection positive
	LO-	*Unused / Opt*	Digital In	Lead-off detection negative
GSR Skin Conductance	VCC	3.3V	Power Rail	Voltage divider configuration
	GND	GND	Common Ground	
	SIG / OUT	GPIO 35	Analog (ADC1_CH7)	Safe ADC1 channel
MQ Gas Sensor (MQ-135)	VCC	5.0V / VBUS	5V Rail	Heater coil requires 5V
	GND	GND	Common Ground	
	AOUT	GPIO 32	Analog (ADC1_CH4)	Resistive divider to match 3.3V scale

3. TCA9548A Multiplexer Channel Map (`Wire` : GPIO 8 / 18)

All devices connected to the multiplexer communicate over the primary I2C bus at **50 kHz**. Only one port is activated at any given moment.

MUX Port	Device Connected	I2C Address	Operational Role	Refresh Strategy
Port 0	*Unused*	—	Reserved for acoustic piezo I2C ADC	—
Port 1	*Unused*	—	Reserved	—
Port 2	**MPU6050**	`0x68`	6-Axis Motion, Chest-Wall Tracking	Polled every $1.2\ \mathrm{s}$
Port 3	**MLX90640**	`0x33`	24×32 Far-Infrared Thermal Sensor	Read at $2\ \mathrm{Hz}$ frame rate
Port 4	**AS7343**	`0x39`	14-Channel Multi-Spectral Optical Sensor	Configured: Gain 128X, $100\ \mathrm{ms}$ int
Port 5	*Unused*	—	Reserved	—
Port 6	*Unused*	—	Reserved	—
Port 7	**SH1106 OLED**	`0x3C`	128×64 Monochrome System Display	Updated every $1.2\ \mathrm{s} - 1.5\ \mathrm{s}$

4. Port Switching & Bus Isolation Functions

To avoid bus contention and driver lockups on the ESP32-S3, all port selections must use the verified shielding wrapper:

```
// Select a specific port on TCA9548A
void tcaSelect(uint8_t port) {
  if (port > 7) return;
  Wire.beginTransmission(0x70);
  Wire.write(1 << port);
  Wire.endTransmission();
}

// Deselect all ports (Isolates all downstream buses)
void tcaDeselect() {
  Wire.beginTransmission(0x70);
  Wire.write(0x00);
  Wire.endTransmission();
}
```

5. Arduino IDE Board & Tool Configurations

To prevent stack overflows, memory alignment faults, and partition errors when flashing the full dual-core build, configure the Arduino IDE as follows:

- **Board:** `ESP32S3 Dev Module`
- **USB CDC On Boot:** `Enabled`
- **CPU Frequency:** `240MHz (WiFi)`
- **Core Debug Level:** `Verbose` (or `None` for production)
- **Flash Mode:** `QIO 80MHz`
- **Flash Size:** `8MB (64Mb)` or `16MB` (depending on DevKit)
- **Partition Scheme:** `Huge APP (3MB No OTA/1MB SPIFFS)`
- **PSRAM:** `OPI PSRAM`
- **Upload Speed:** `921600`